ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **WINTER SEMESTER, 2024–2025**

DURATION: 2 HOURS **FULL MARKS: 120**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. A system of linear equations is given in matrix form as $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 \\ 2 & 2 & 12 & -2 & -4 \\ 3 & 4 & 15 & -4 & -5 \\ 1 & 4 & -3 & -3 & 3 \\ -2 & -4 & -6 & 0 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

a) Between $\mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -1 \\ -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -1 \\ 3 \\ 2 \\ -8 \\ 10 \end{bmatrix}$, for which vector $\mathbf{b}$ is the system of linear equations solvable? Provide a justification for your answer. 12 (CO2) (PO1)

b) Find the complete solution to the system $A\mathbf{x} = \mathbf{b}$ by addressing the following: $1+2+4+1+1$ (CO2) (PO1)

i. Identify the **pivot variables** and the **free variables** based on the row-reduced form of the matrix A.

ii. Determine the **particular solution** ($\mathbf{x}_p$) for the vector $\mathbf{b}$ from Question 1.a for which a solution exists.

iii. Determine the **special solution(s)** ($\mathbf{x}_s$) that form the basis of the nullspace.

iv. Express the complete solution as $\mathbf{x}_c = \mathbf{x}_p + \sum c_i \mathbf{x}_{s,i}$, where c_i are the free variables (arbitrary constants) and $\mathbf{x}_{s,i}$ are the special solutions.

v. State the nature of the solution (whether it is always solvable, has a unique solution, or has infinitely many solutions).

c) Is the solution obtained in Question 1.b)iii a vector space? If yes, justify your answer. If no, provide a counterexample to explain why. 3 (CO1) (PO1)

d) The vectors $\mathbf{v}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -10 \\ 4 \\ 0 \\ 4 \\ 1 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} -2 \\ -2 \\ 2 \\ 2 \\ 1 \end{bmatrix}$ cannot form a basis for the Left Nullspace of A, $N(A^T)$. $5+5$ (CO2) (PO1)

i. Provide at least two independent arguments to justify why these vectors cannot represent a basis for $N(A^T)$.

ii. It is guaranteed that one or more vectors from the above set do belong to $N(A^T)$. Determine which of the provided vector(s) form a basis for $N(A^T)$, and justify your answer.

e) Based on your findings in Question 1.b and Question 1.d, answer the following questions regarding the Columnspace $C(A)$, Nullspace $N(A)$, and Left Nullspace $N(A^T)$ of matrix A. For all **three subspaces**: 3 + 3 (CO1) (PO1)

i. Write down the **bases** for each of the three subspaces.

ii. State the **dimension** of each subspace and identify the **vector space** (e.g., $\mathbb{R}^n$) in which it resides.

2. Figure 1 illustrates the inflow and outflow of a network containing three nodes: X, Y, and Z. Figure 1 illustrates the inflow and outflow of a network containing three nodes: X, Y, and Z. The weights of the directed edges between each pair of nodes are equal. For example, for the nodes X and Y,

$$2a + 7c = 3b - 11$$

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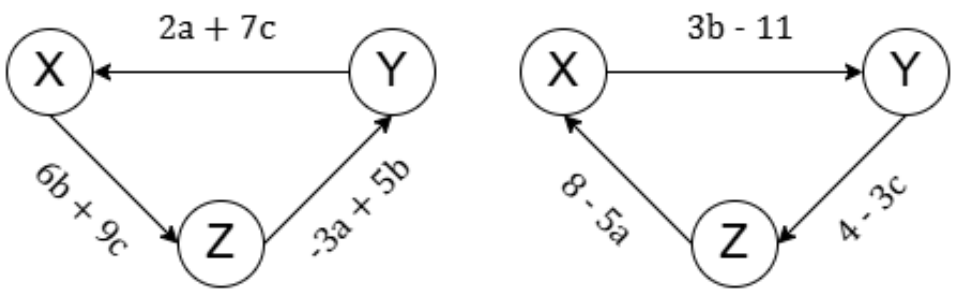

Figure 1: Inflow and Outflow Diagram of a Graph

a) Determine the values of the unknown flows a, b, and c by completing the following steps: 2 + 1 + 7 + 1 (CO2) (PO1)

i. Derive the system of three linear equations based on the flow conservation principle at nodes X, Y, and Z as shown in Figure 1.

ii. Express the system of equations (obtained in Question 2.a)i) in the matrix form $A\mathbf{x} = \mathbf{b}$, clearly identifying the coefficient matrix A, the variable vector $\mathbf{x}$, and the constant vector $\mathbf{b}$.

iii. Perform Gaussian Elimination on the augmented matrix $[A|\mathbf{b}]$. You must show the specific elimination matrices E_{ij} used to zero out the entry at row i and column j. If a pivot is zero, utilize a permutation matrix P_{ij} to perform a row swap.

iv. Determine the final values of a, b, and c using back-substitution on the resulting upper triangular matrix.

b) Factorize the coefficient matrix A from Question 2.a)ii into LDU form, where L is a unit lower triangular matrix (the inverse of the product of elimination matrices, E^{-1}), D is a diagonal matrix containing the pivots, and U is the unit upper triangular matrix obtained by scaling the rows of the matrix from Question 2.a)iii so that all diagonal entries are 1. 6 + 3 (CO1) (PO1)

i. Determine the specific entries for matrices L, D, and U.

ii. Calculate the **trace** and the **determinant** for each of the three matrices: L, D, and U.

3. a) A linear transformation maps the vector $\begin{bmatrix} -8 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} -8 \\ 4 \end{bmatrix}$. The transformation compresses the entire space by a factor of half and also reverses its orientation. The trace of the transformation matrix is 1. 10 (CO2) (PO1)

Find the transformation matrix corresponding to this transformation. In other words, determine the basis vectors (columns) of the transformation matrix.

b) Let $\vec{p}$, $\vec{q}$, and $\vec{r}$ be three linearly independent vectors. The following regions are defined: 4 + 10 (CO2) (PO1)

$$X = \{ k_1\vec{p} + k_2\vec{q} + \vec{r} \mid k_1, k_2 \in \mathbb{R} \} \quad \text{and} \quad Y = \{ k_3\vec{r} \mid k_3 \in \mathbb{R} \},$$

where k_1, k_2, and k_3 are scalars.

i. Roughly sketch X and Y.

ii. Determine whether the set $X \cup Y$ and the set $X \cap Y$ form vector spaces. Provide proper justification for your answers.

c) Show that for any permutation matrix P, the transpose of P is equal to its inverse: 6 (CO1) (PO1)

$$P^T = P^{-1}$$

Your argument should be based on the defining properties of permutation matrices and the effect of transposition and inversion on them.

4. a) There are n students at a renowned International University. Their main occupation is forming various societies every month so that students can become Presidents, Secretaries, or occupy made up positions on impressive sounding Executive Committees. After a while, this started to get boring, and the university began losing its aura. In order to limit the number of societies, the *ExcludedFromEC* Council decreed the following innocent-looking rules: 5 + 10 (CO2) (PO1)

- Each society has to have an **odd number** of members.
- Every **pair of societies** must have an **even** number of members in **common**.

Show that under these rules, it is impossible to form more societies than n, the number of students. Attempt to solve this problem by approaching it in the following way:

i. Let the number of students be n (labeled $1, 2, \dots, n$), and let there be m societies $S_1, S_2, \dots, S_m$. Construct an $m \times n$ incidence matrix $A = (a_{ij})$, where

$$a_{ij} = \begin{cases} 1, & \text{if } j \in S_i \text{ (Student } j \text{ is in Society } i\text{)}, \\ 0, & \text{otherwise.} \end{cases}$$

(In other words, construct a matrix with societies along rows and students along columns, if a student belongs to a society put 1, otherwise 0). Construct this matrix for small values of n and m of your choice, making sure that $n > 3$, $m > 3$, and $n \neq m$.

ii. Discuss your solution approach. What would be sufficient to show that it is not possible to form more than n societies under the new rules? If possible, provide the proof.

b) Given, 7 (CO1) (PO1)

$$S = \left\{ \begin{bmatrix} x^3 \\ y^9 \end{bmatrix} \middle| \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \right\}$$

Is S a vector space? If yes, justify your answer. If no, provide a counterexample to demonstrate why it is not.

c) There are m rows and n columns in a matrix M. The dimension (number of basis vectors) of its nullspace is $n - r$. 3 + 5 (CO1) (PO1)

i. What is the span of the columns of the stated matrix M? (e.g., nothing, line, plane, or specific vector space $\mathbb{R}^k$).

ii. If the matrix is square and its rank is the same as the number of columns, describe the nullspace of the matrix and comment on the span of its columns.